

W3WI_110.2 - Verteilte Systeme

Dozent:
Kontakt:
Version:

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LectureDoc2
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1.0 (23EG/EH)
Developed by M. Eichberg

KI Verwendung:

Mouse Navigation

When you click on the left side of a slide you'll go to the previous slide. When you click on the right half the next step will be done; i.e., the next animation will be started or you'll advance to the next slide.

Touch Support

Slide View

Swipe Gesture - Left and Right

You can swipe left or right to trigger the next animation step or to undo the last step.

Swipe Gesture - Up and Down

You can swipe up or down to go to the previous or next slide.

Pinch Gesture

You will go back to the document view.

Document View

Zoom Gesture

You will change to the slide view and the slide belonging to the section will be shown.

Keyboard Bindings

Key	Action
<i>Navigation</i>	
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←	undo last step
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b	toggle black screen
r	reset animation progress for current slide
r (5 times in a row)	reset all animations
<i>Presentation</i>	
w	open secondary (linked) window

Kerninhalte gem. MHB

- Terminologie, Konzepte, Architekturen, Anforderungsprofile und Architekturmodelle für verteilte Systeme
- Entwurfs- und Implementierungsansätze
- Vergleich unterschiedlicher Middleware-Konzepte
- Synchron und asynchrone Kommunikation, entfernter Methodenaufruf
- Asynchrone Kommunikation und Messaging-Systeme
- Sicherheit in verteilten Systemen

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Prüfungsleistung - Portfolio

Hintergrund

- das Modul hat 55 VL
- Verteilte Systeme hat 22VL
- Die verteilte Systeme geht mit 50 von 120 Modulpunkten ein

- 2 Bestandteile:

01 Vorträge - max. 15 Punkte - 20 Minuten pro Person

02 Programmieraufgabe - max. 35 Punkte

▲ **Achtung!** Wenn Sie einen Vortrag halten, sollten bei der Programmieraufgabe nicht im selben Team sein.

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Vorträge - Rahmenbedingungen

- Die Präsentationen sollen sich insbesondere mit den Kerninhalten der Vorlesung beschäftigen und insbesondere konzeptioneller Natur sein.

D. h. nach der Darstellung des Anwendungszweckes gilt es die Architektur darzustellen, wie mit Fehlern umgegangen wird, welche Services angeboten werden, welche Garantien/Sicherheitsaspekte umgesetzt werden, wie wird die Skalierbarkeit erreicht, etc.

Keine Werbevorträge!

- Die Präsentationen sind am Abend vor dem ausgemachten Termin hochzuladen in Moodle.

- Die Vorträge dürfen während des Vortrags nicht mehrfach rotieren.

D. h. gefordert ist, dass erst der erste Vortragende seinen Teil vollständig vorträgt, dann der zweite, usw.. Dies ist für die Notenförmigkeit entscheidend.

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Vorträge - Themen

Themen, die zu besetzen sind

Thema	Anzahl Stud.
Byzantine Faults 	1 Stud.
LDAP 	2 - 3 Stud.
Backend Subsetting and connection churn 	1 Stud.
Virtualization Ziel dieser Präsentation ist es, einen Überblick über die verschiedenen Virtualisierungstechnologien zu geben und die Gemeinsamkeiten und Unterschiede zwischen ihnen zu erläutern. Es ist auch möglich, die Unterschiede anhand konkreter Produkte zu demonstrieren. ausgewählte Schlüsselwörter: Containers (z. B., Docker, Linux), Hypervisor, Hypervisors (KVM, Xen, Hyper-V)	3 - 5 Stud.
Real time 	2 Stud. (mit Interesse an Linux)
HTTP/3 and QUIC or HTTP over QUIC 	2 Stud.
Themen, die vorgetragen werden können Swipe Gesture - Left and Right You can swipe left or right to trigger the next animation step or to undo the last step. Swipe Gesture - Up and Down You can swipe up or down to go to the previous or next slide. Pinch Gesture You will go back to the document view. Document View Zoom Gesture You will change to the slide view and the slide belonging to the section will be shown.	2 Stud. 2 Stud. 2 Stud. 2 Stud. 4 Stud.
Neo4J 	2 Stud.

Keyboard Bindings

 connection churn =  Verbindungsabbruch

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Vorträge - Datum

Pro Termin werden Vorträge von insgesamt 100 Minuten Länge gehalten.

VORLÄUFIGE PLANUNG

Date	Topic
9. April	1. (Byzantine faults), 4. (Virtualization), 2. (LDAP), 5. (Monitoring)
16. April	7. (Ceph), 11. (Testing), 12. (Neo4j), 13. (Docker Swarm vs. Kubernetes)
23. April	8. (Raft), 9. (Gossip), 3. (Connection Churn), 6. (HTTP/3), 10. (gRPC)

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